

Optimization

Project 3 – Non-Linear Programming

Deliverables

One Python code file (.py or .ipynb) and one well-written PDF file, submitted to Canvas. Your report should go into some detail about how you solved the problem, include some graphs that explain your results, and include relevant code chunks in the final output. 66% of your grade will be based on whether you get the problem right or not, the remaining 34% will be based on the quality of your analysis. We will re-run your code with a new data set. If you don't get the right answer or your Python file doesn't run, we will go through your code and give partial credit accordingly. The easier it is to read your code the easier it is for us to understand what you're doing, so use a lot of comments in your code!

Problem Overview

The newsvendor model is a common model used in operations. Let's say we work for a manufacturing company; if we do not know exactly what the demand of a product will be, how many should we manufacture? This is called the newsvendor model because we typically assume that if you produce too much you have to throw away the excess production – nobody wants to buy yesterday's newspaper! In this project you will solve a few extensions of the standard newsvendor model, but let's first talk about the standard NVM.

We will assume that demand is random, but follows some distribution. After we make our production decision, the random demand will reveal itself. If we produced more than the demand, then we only sell what was demanded, but if we produced less than the demand we only sell what we produced. Our job is to decide what quantity, q , to produce that maximizes the expected profit:

$$\max_q E[p \min(q, D) - qc]$$

where p is the price at which we sell our product, D is the random demand for the product, and c is the cost to manufacture our product. Usually when we solve this problem we do not use the true distribution of D , but rather data from D . Suppose we have several observations of demand, D_i , over n days of data, then we would rewrite our objective as:

$$\max_q \frac{1}{n} \sum_{i=1}^n (p \min(q, D_i) - qc).$$

This is a non-linear program (the $\min(q, D_i)$ term makes this non-linear), but if we follow a similar trick as we did in project 2, we can reformulate this as a linear program: the profit is always either $pD_i - qc$ or $pq - qc$, whichever is smaller. Let's create some dummy variables, h_i , that either takes on the value $pD_i - qc$ or $pq - qc$, and then average the h 's. We can rewrite the objective as

$$\begin{aligned} & \max_{q, h} \frac{1}{n} \sum_{i=1}^n h_i \\ \text{s.t.} \quad & h_i \leq pD_i - qc \\ & h_i \leq pq - qc \\ & h_i \geq -\infty \end{aligned}$$

This is then a linear program that can be solved with gurobi, if you have demand data. We will now consider extensions of this model.

In our first extension, we will assume that if you don't print enough newspapers to satisfy demand, then you can send a rush order to the printers to print all that you need. These rushed newspapers cost more to print, we will call the cost per rushed newspaper g , where $g > c$. Additionally, if you print more than the demand you must pay a disposal fee of t dollars per newspaper. You could alternatively think about this as receiving money for taking excess newspapers to a recycling center if $t < 0$, but for now we'll think of $t > 0$. Putting this together, the objective function is:

$$\max_q \frac{1}{n} \sum_{i=1}^n (pD_i - qc - g(D_i - q)^+ - t(q - D_i)^+)$$

where $(x)^+ = \max(x, 0)$. Part of this project will be to reformulate this as a linear program.

In our second extension of the NV model we will assume that price impacts demand linearly with error, $D_i = \beta_0 + \beta_1 p + \epsilon_i$, and jointly solve for the optimal price and quantity to print. This is in the model with disposal costs, and quick printing costs. To do this, you will be given a data set with price and demand data. You will fit a linear regression to this data, and you will assume the source of randomness is the residuals of this regression. This means once you find β_0 and β_1 you can find lots of demand data points for any one price: get all your residuals and plug them into the regression equation for that one price. That is, your data becomes just the residuals of your regression, instead of the original price/demand data. If you find $\beta_0 = 1000$ and $\beta_1 = -2$ and two of the residuals are 15 and -9, then for a price of \$2 you get 2 demand data points of $1000 - 2 \cdot 2 + 15 = 1011$ and $1000 - 2 \cdot 2 - 9 = 998$.

You can see that when demand becomes a function of price like this you must multiply pD_i but this becomes $p(\beta_0 + \beta_1 p + \epsilon_i)$. This is then a quadratic function of p .

To solve this problem, you will have to create some dummy variables for each day, h_i , that represent some part of the objective, in the original NVM we let h_i represent the entire profit for each day. I can think of 2 ways to reformulate these as something that gurobi can solve. First, stick with what we did above where h_i is the profit on each day. Then the objective just becomes the average of the h_i 's and you have 2 constraints for each day's profit, and those constraints become quadratic when price impacts demand. This would be a Quadratically Constrained Program (QCP).

Alternatively, you could let h_i just be the negative of cost on each day. Then the objective is the average of revenue plus negative cost, where the revenue part comes from pD_i and the negative cost comes from h_i . This way the quadratic part would stay in the objective instead of going down to a constraint when price impacts demand. You still need 2 constraints for the negative cost dummy variables, but these are linear constraints instead of quadratic. This method is a little trickier to formulate but then the problem would be a QP instead of a QCP, which is way easier for gurobi to solve. The reason we would define h_i to be negative cost is that we're maximizing over h , so we want negative costs to be as big as possible; we don't want big costs. Also, the constraints need to be upper bounds on h , not lower bounds. Remember that costs are always positive in this model, so negative costs are always negative. Set your lb accordingly.

Specifics

- 1) Data on price and demand are included in the assignment. Fit a linear regression model to this data set.
- 2) Let $c=0.5$, $g=0.75$, and $t=0.15$. Using the residuals, assume the price is $p=1$ and generate demand data.
- 3) Solve the optimal quantity to produce when $p=1$. This is not quadratic at all, just an LP.

- 4) Now let price impact demand and solve either the resulting QCP or QP. What are the optimal price and quantity to print?
- 5) If you get in a bind and can't figure out how to formulate the QP/QCP you can formulate it as a general NLP and solve it using `scipy.optimize`, but that will result in a 10 point deduction from the project grade.
- 6) We are now interested to know how sensitive the optimal price and quantity are to our data set. Take a bootstrap sample of the original dataset. Go back and fit new beta's to the new bootstrapped dataset and redo step 4. Find the optimal price and quantity.
- 7) Repeat this process of getting new bootstrapped data, fitting new betas to the newly simulated data and finding the optimal price/quantity many times. Make histograms of the optimal price and quantity. Make a scatterplot with histograms on the x and y axis for this. Make a histogram of expectation of profits.
- 8) You work for a publishing company. Your boss has been using the standard NV model to set printing decisions using the dataset given. Is your boss' model as good as this one? Could you increase revenue by switching models for production/pricing decisions? Describe the advantages and disadvantages of both techniques. Your boss is pretty technical and understands optimization, so don't be afraid to include quantitative material. Your boss is also busy, so be sure to include some visualizations to get the important points across. For the purpose of your report, you can assume that your boss is interested in the data posted with the project. Write this as if you were going to give it to your boss as a technical report that may be shared with the CEO. That means, make it look GOOD!